Chapter 16 Notes

Section 1

Theorem 2.7

Here is a proof of the classical part of this theorem. Consider some model modal $\mathcal{M}$ for which the hypotheses of the problem hold

Atomic case: In this case X is atomic. Thus it must be that $A = X$ and $Y = B$, meaning, because we have assumed $A \equiv B$ is valid in $\mathcal{M}$, $X \equiv Y$ (which is the same formula) is also valid in $\mathcal{M}$.

Negation case: In this case X is of the form $\sim\phi$. Then either $A = X$ or A is a proper subformula of ϕ. In the first case, obviously it follows that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, Y must be of the form $\sim\psi$ (where $\phi \equiv \psi$ is valid over $\mathcal{M}$ by assumption). Then $\phi \supset \psi$ is valid over $\mathcal{M}$ (by assumption), meaning $\sim\psi \supset \sim\phi$ is valid over $\mathcal{M}$. Similarly, $\sim\phi \supset \sim\psi$ is valid over $\mathcal{M}$ (because $\psi \supset \phi$ is valid over $\mathcal{M}$). Thus $\sim\phi \equiv \sim\psi$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Conjunction case: In this case X is of the form $\phi \wedge \psi$. Then either $A = X$ or A is a subformula of either ϕ or ψ. In the first case, obviously it follows that $X \equiv Y$ is valid in $\mathcal{M}$. In the second case, Y must be of the form $\theta \wedge \gamma$, where both $\theta \equiv \phi$ and $\gamma \equiv \psi$ are true in $\mathcal{M}$. Then $(\phi \wedge \psi) \equiv (\theta \wedge \gamma)$, that is, $X \equiv Y$, being valid in $\mathcal{M}$, follows easily.

Existential case: In this case X is of the form $(\exists x)\phi(x)$. Then either $A = X$ or A is a subformula of $\phi(x)$. Again, the first case is obvious. In the second case, we show that (1) $(\exists x)\phi(x) \supset (\exists x)\psi(x)$ is valid over $\mathcal{M}$. So, for any world p of the model, suppose that $(\exists x)\phi(x)$ is true at p. Then $\phi(a)$ is true at p for some $a \in \mathcal{D}$. Using the induction assumption, it is easy to see that this means $\psi(a)$ must be true at p (because $A \equiv B$ is assumed to be valid over $\mathcal{M}$). Thus $(\exists x)\psi(x)$ is true at p. Hence, because (1) is true at all possible worlds of $\mathcal{M}$, it follows that it is valid over $\mathcal{M}$. Similar reasoning shows that the converse of (1) is valid over $\mathcal{M}$. Hence $(\exists x)\phi(x) \equiv (\exists x)\psi(x)$, that is, $X \equiv Y$, is valid over $\mathcal{M}$.

Exercise 2.1

(1) $p \vDash (X \vee Y)$ iff $p \vDash \sim(\sim X \wedge \sim Y)$ (by definition of $\vee$) iff $p \nvDash (\sim X \wedge \sim Y)$ iff either $p \nvDash \sim X$ or $p \nvDash \sim Y$ iff either $p \vDash X$ or $p \vDash Y$.

(2) $p \vDash (X \supset Y)$ iff $p \vDash \sim(X \wedge \sim Y)$ (by definition of $\supset$) iff $p \nvDash (X \wedge \sim Y)$ iff either $p \nvDash X$ or $p \nvDash \sim Y$ iff either $p \nvDash X$ or $p \vDash Y$.

(3) $p \vDash (\forall x)\phi(x)$ iff $p \vDash \sim(\exists x)\sim\phi(x)$ (by definition of $\forall$) iff $p \nvDash (\exists x)\sim\phi(x)$ iff $p \nvDash \sim\phi(d)$ for each $d \in \mathcal{D}$ iff $p \vDash \phi(d)$ for each $d \in \mathcal{D}$.

(4) $p \vDash \Diamond X$ iff $p \vDash \sim\Box\sim X$ (by definition of $\Diamond$) iff $p \nvDash \Box\sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \nvDash \sim X$ iff for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$.

Exercise 2.2

(1) We first prove the validity of the K axiom over every modal model (note that the converse is not valid). Consider some modal model $\mathcal{M}$, and one of its possible worlds p. We must show that $p \vDash \Box(X \supset Y) \supset (\Box X \supset \Box Y)$. It suffices to show

that, if $p \vDash \Box(X \supset Y)$, then it follows that $p \vDash (\Box X \supset \Box Y)$. So suppose $p \vDash \Box(X \supset Y)$. Then this means that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (X \supset Y)$. It suffices to show that, if $p \vDash \Box X$, then it follows that $p \vDash \Box Y$. So suppose that $p \vDash \Box X$. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash X$. But also $q \vDash (X \supset Y)$, meaning $q \vDash Y$. Thus $p \vDash \Box Y$. Hence $p \vDash (\Box X \supset \Box Y)$ holds, and consequently, $p \vDash \Box(X \supset Y) \supset (\Box X \supset \Box Y)$ also holds, and thus $\Box(X \supset Y) \supset (\Box \supset \Box Y)$ is valid.

(2) We now prove the validity of the Barcan formula over every modal model. Consider some modal model $\mathcal{M}$, and one of its possible worlds p. We must show that $p \vDash (\forall x)\Box\phi(x) \supset \Box(\forall x)\phi(x)$. It suffices to show that, if $p \vDash (\forall x)\Box\phi(x)$, then it follows that $p \vDash \Box(\forall x)\phi(x)$. So suppose that $p \vDash (\forall x)\Box\phi(x)$. Thus $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$, meaning $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. To show that $p \vDash \Box(\forall x)\phi(x)$ holds, we must show that, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$. So consider some such q. We must show that $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. But we have already shown this. Thus $q \vDash (\forall x)\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x)$, and consequently, the Barcan formula is valid.

Note that $(\forall x)\Diamond\phi(x) \supset \Diamond(\forall x)\phi(x)$ is not valid.

Exercise 2.3

We show that the converse of the Barcan formula, that is, $\Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, is valid over every modal model. Consider any modal model $\mathcal{M}$, and one of its possible worlds p. It suffices to show that, if $p \vDash \Box(\forall x)\phi(x)$ holds, then $p \vDash (\forall x)\Box\phi(x)$ also holds. So suppose that $p \vDash \Box(\forall x)\phi(x)$ holds. Then, for every $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (\forall x)\phi(x)$ holds, meaning $q \vDash \phi(d)$ holds for each $d \in \mathcal{D}$. To show that $p \vDash (\forall x)\Box\phi(x)$ holds, we must show that $p \vDash \Box\phi(d)$ holds for each $d \in \mathcal{D}$. So consider some such d. Then we must show that $q \vDash \phi(d)$ holds for each $q \in \mathcal{G}$ with $p\mathcal{R}q$. So consider some such q. But we already know that $q \vDash \phi(d)$ holds. Thus $p \vDash \Box\phi(d)$, meaning $p \vdash (\forall x)\Box\phi(x)$, meaning $p \vDash \Box(\forall x)\phi(x) \supset (\forall x)\Box\phi(x)$, and thus the converse of the Barcan formula is valid.

Note that $\Diamond(\forall x)\phi(x) \supset (\forall x)\Diamond\phi(x)$ is also valid.

Exercise 2.4

For this, we first prove by induction (using Exercise 2.3) that

$$(\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n) \equiv \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$$

is valid. The base case is just Exercise 2.3. Now suppose the result holds for some n. Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \equiv \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid, meaning we must show that both

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

and

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

are valid.

On the one hand, we show that

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid. So suppose

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

Then, by Exercise 2.3 and the Replacement Theorem,

$$p \vDash (\forall x_1)...(\forall x_n)\Box(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

meaning, by the induction assumption,

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

Hence

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1}) \supset \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

is valid.

On the other hand, we show that

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

is valid. So suppose that

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

Then

$$p \vDash (\forall x_1)\Box(\forall x_2)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1})$$

by Exercise 2.3, and then

$$p \vDash (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

by the induction assumption and the Replacement Theorem. Hence

$$p \vDash \Box(\forall x_1)...(\forall x_{n+1})\phi(x_1, ..., x_{n+1}) \supset (\forall x_1)...(\forall x_{n+1})\Box\phi(x_1, ..., x_{n+1})$$

is valid.

Thus we know that

$$(\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n) \equiv \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$$

is valid. Then $p \vDash \Box\phi(x_1, ..., x_n)$ iff $p \vDash (\forall x_1)...(\forall x_n)\Box\phi(x_1, ..., x_n)$ iff $p \vDash \Box(\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$ iff $q \vDash (\forall x_1)...(\forall x_n)\phi(x_1, ..., x_n)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$ iff $q \vDash \phi(x_1, ..., x_n)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$.

Exercise 2.5

(a) We show that $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ is valid over every modal model. So consider some modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash \Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$. It suffices to show that, if $p \vDash \Box(X \wedge Y)$, then $p \vDash (\Box X \wedge \Box Y)$. So suppose that $p \vDash \Box(X \wedge Y)$. Thus $q \vDash (X \wedge Y)$ holds for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. This means that $q \vDash X$ and $q \vDash Y$ both hold for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, hence both $p \vDash \Box X$ and $p \vDash \Box Y$ hold, and consequently, $p \vDash (\Box X \wedge \Box Y)$ holds. Thus $p \vDash \Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$, meaning $\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$ is valid over $\mathcal{M}$, hence valid over every modal model.

(b) We show that $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ is valid over every modal model. So consider some modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash (\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$. It suffices to show that, if $p \vDash (\Box X \wedge \Box Y)$, then $p \vDash \Box(X \wedge Y)$. So suppose that $p \vDash (\Box X \wedge \Box Y)$. Then $p \vDash \Box X$ and $p \vDash \Box Y$. Hence $q \vDash X$ and $q \vDash Y$ both hold for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. This means $q \vDash (X \wedge Y)$ holds for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. But this means $p \vDash \Box(X \wedge Y)$ holds, and thus $p \vDash (\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$, meaning $(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$ is valid over $\mathcal{M}$, hence valid over every modal model.

(c) We show that $(\Diamond X \vee \Diamond Y) \equiv \Diamond(X \vee Y)$ is valid over every modal model. Consider any modal model $\mathcal{M}$, and any of its possible worlds p.

On the one hand, we show that $p \vDash (\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$. It suffices to show that, if $p \vDash (\Diamond X \vee \Diamond Y)$, then $p \vDash \Diamond(X \vee Y)$. So suppose that $p \vDash (\Diamond X \vee \Diamond Y)$. Thus either $p \vDash \Diamond X$ or $p \vDash \Diamond Y$, hence (1) there is either some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash X$, or there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash Y$. To show that $p \vDash \Diamond(X \vee Y)$ holds, we must show that there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash (X \vee Y)$. But, we are guaranteed to have such a q by (1). Hence $p \vDash \Diamond(X \vee Y)$ holds, meaning $p \vDash (\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$, and thus $(\Diamond X \vee \Diamond Y) \supset \Diamond(X \vee Y)$ is valid over all modal models.

On the other hand, we show that $p \vDash \Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$. It suffices to show that, if $p \vDash \Diamond(X \vee Y)$, then $p \vDash \Diamond(X \vee Y)$. So suppose that $p \vDash \Diamond(X \vee Y)$. Then, for some $q \in \mathcal{G}$ with $p\mathcal{R}q$, $q \vDash (X \vee Y)$, meaning either $q \vDash X$ or $q \vDash Y$. But this means that there is either some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash X$, or there is some $q \in \mathcal{G}$ with $p\mathcal{R}q$ such that $q \vDash Y$, meaning either $p \vDash \Diamond X$ or $p \vDash \Diamond Y$, and thus $p \vDash (\Diamond X \vee \Diamond Y)$. Hence $p \vDash \Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$, meaning $\Diamond(X \vee Y) \supset (\Diamond X \vee \Diamond Y)$ is valid over all modal models.

(d) We first show that $(\Diamond X \wedge \Diamond Y) \supset \Diamond(X \wedge Y)$ is not valid (note that the converse is valid). Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that it could be the case that $p \nvDash (\Diamond X \wedge \Diamond Y) \supset \Diamond(X \wedge Y)$, which is the same as saying that it could be the case that $p \vDash (\Diamond X \wedge \Diamond Y) \wedge \sim\Diamond(X \wedge Y)$. Well, it certainly could be the case that

$p \vDash (\Diamond X \wedge \Diamond Y)$. If it is, then there is some q accessible from p such that $q \vDash X$, and there is some r accessible from p such that $r \vDash Y$. To conclude that $p \vDash \Diamond(X \wedge Y)$, we would need to know that there is some p' accessible from p such that $p' \vDash (X \wedge Y)$. But, because it could be that $q \neq r$, we are not guaranteed this.

Second, we show that $\Box(X \vee Y) \supset (\Box X \vee \Box Y)$ is not valid. Consider any modal model $\mathcal{M}$ and any of its possible worlds p. We must show that it could be the case that $p \nvDash \Box(X \vee Y) \supset (\Box X \vee \Box Y)$, which is the same as saying that it could be the case that $p \vDash \Box(X \vee Y) \wedge \sim(\Box X \vee \Box Y)$. Well, it certainly could be the case that $\Box(X \vee Y)$. If it is, then $q \vDash (X \vee Y)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, meaning for each such q, either $q \vDash X$ or $q \vDash Y$. But this does not mean that either $q \vDash X$ holds for all such q, or $q \vDash Y$ holds for all such q, which is what we would need for $p \vDash (\Box X \vee \Box Y)$. Hence we cannot necessarily conclude $p \vDash (\Box X \vee \Box Y)$, meaning $(\Box X \vee \Box Y)$ is not valid.

Exercise 2.6

We show that $\Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$ is valid in every modal model (note that the converse is not valid). Consider any modal model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash \Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$. It suffices to show that, if $p \vDash \Box(X \equiv Y)$, then $p \vDash (\Box X \equiv \Box Y)$. So suppose that $p \vDash \Box(X \equiv Y)$. Thus (1) $q \vDash (X \equiv Y)$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$. To show that $p \vDash (\Box X \equiv \Box Y)$, we must show that $p \vDash (\Box X \supset \Box Y)$, and $p \vDash (\Box Y \supset \Box X)$.

On the one hand, suppose that $p \vDash \Box X$. Thus $q \vDash X$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, but this also means (by (1)) that $q \vDash Y$ for all $q \in \mathcal{G}$ with $p\mathcal{R}q$, and thus $p \vDash \Box Y$.

The case for showing $p \vDash (\Box Y \supset \Box X)$ is similar to the first case.

Thus we have both $p \vDash (\Box X \supset \Box Y)$ and $p \vDash (\Box Y \supset \Box X)$, meaning $p \vDash (\Box X \equiv \Box Y)$, which is what we were trying to prove.

Exercise 2.7

For this exercise, "Classical logic" as a reason in the derivation stands for something derived due to the assumption that the axiom system we are working in contains the standard rules for classical logic.

(a)

1.	$((X \wedge Y) \supset X)$	(Classical logic)
2.	$\Box((X \wedge Y) \supset X)$	(1, Necessitation rule)
3.	$\Box(X \wedge Y) \supset \Box X$	(2, K axiom)

(b) For this, note that part (a) can of course be altered to prove that $\Box(X \wedge Y) \supset \Box Y$.

1.	$\Box(X \wedge Y) \supset \Box X$	(Part (a))
2.	$\Box(X \wedge Y) \supset \Box Y$	(Part (a))
3.	$\Box(X \wedge Y) \supset (\Box X \wedge \Box Y)$	(1, 2, Classical logic)

(c)

1.	$(X \supset (Y \supset (X \wedge Y)))$	(Classical logic)
2.	$\Box(X \supset (Y \supset (X \wedge Y)))$	(1, Necessitation rule)
3.	$\Box X \supset \Box(Y \supset (X \wedge Y))$	(2, K axiom)
4.	$\Box X \supset (\Box Y \supset \Box(X \wedge Y))$	(3, K axiom)
5.	$(\Box X \wedge \Box Y) \supset \Box(X \wedge Y)$	(4, Classical logic)

Section 3

Notes

I was unable to prove the $S4$ completeness of the system spoken of in this section (I tried using modal tableau, but was unable to handle the $\Diamond$ case), but here are some proofs in the system of some of the $S4$ valid formulas discussed in this chapter.

Let "Ref." stand for reflexitivity axiom, "Trans." stand for transitivity axiom, "K" stand for K axiom, and "Nec." stand for Necessitation rule.

$X \supset \Diamond X$

1.	$\Box\sim X \supset \sim X$	(Ref.)
2.	$\sim\sim X \supset \sim\Box\sim X$	(1, Classical logic)
3.	$X \supset \Diamond X$	(2, Classical logic)

$\Diamond\Diamond X \supset \Diamond X$

1.	$\Box\sim X \supset \Box\sim X$	(Classical logic)
2.	$\Box\sim X \supset \sim\sim\Box\sim X$	(1, Classical logic)
3.	$\Box(\Box\sim X \supset \sim\sim\Box\sim X)$	(2, Nec.)
4.	$\Box\Box\sim X \supset \Box\sim\sim\Box\sim X$	(3, K)
5.	$\Box\sim X \supset \Box\Box\sim X$	(Ref.)
6.	$\Box\sim X \supset \Box\sim\sim\Box\sim X$	(4, 5, Classical logic)
7.	$\sim\Box\sim\sim\Box\sim X \supset \sim\Box\sim X$	(6, Classical logic)
8.	$\Diamond\Diamond X \supset \Diamond X$	(7, Def. of $\Diamond$)

$\Diamond X \supset \Diamond\Diamond X$

1.	$\Box\sim X \supset \sim X$	(Ref.)
2.	$\sim\sim\Box\sim X \supset \sim X$	(1, Classical logic)
3.	$\Box(\sim\sim\Box\sim X \supset \sim X)$	(2, Nec.)
4.	$\Box\sim\sim\Box\sim X \supset \Box\sim X$	(3, K)
5.	$\sim\Box\sim X \supset \sim\Box\sim\sim\Box\sim X$	(4, Classical logic)
6.	$\Diamond X \supset \Diamond\Diamond X$	(5, Def. of $\Diamond$)

From the previous two proofs, we have proven $\Diamond\Diamond X \equiv \Diamond X$, and thus this is an alternative proof of one part of Exercise 3.4.

$\Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$

1.	$\Box(X \equiv Y) \supset \Box((X \supset Y) \wedge (Y \supset X))$	(Def. of $\equiv$)
2.	$\Box(X \equiv Y) \supset (\Box(X \supset Y) \wedge \Box(Y \supset X))$	(1, Exercise 2.7(b))
3.	$\Box(X \equiv Y) \supset ((\Box X \supset \Box Y) \wedge (\Box Y \supset \Box X))$	(2, K)
4.	$\Box(X \equiv Y) \supset (\Box X \equiv \Box Y)$	(3, Def. of $\equiv$)

Exercise 3.1

(1) First we show that $\Box(\Box A \supset \Box A)$ is $S4$ valid. So consider any $S4$ model $\mathcal{M}$, and any of its possible worlds p. We must show that $p \vDash \Box(\Box A \supset \Box A)$. It suffices to show that $q \vDash (\Box A \supset \Box A)$ for each q accessible at p. So consider any such q. It suffices to show that, if $q \vDash \Box A$, then $q \vDash \Box A$. But this is obvious. Hence $p \vDash \Box(\Box A \supset \Box A)$, and, consequently, $\Box(\Box A \supset \Box A)$ is $S4$ valid (indeed, it is valid over all modal models).

Alternatively, this follows by part (1) of Proposition 2.6 and the fact that $p \vDash (\Box A \supset \Box A)$ is valid over all modal models.

(2) Next we show that $\Box(\Box A \vee \Box{\sim}\Box A)$ is not $S4$ valid. So consider any $S4$ model $\mathcal{M}$, and any of its possible worlds p. Well, it certainly could be that $q \nvDash A$ for some q accessible at p, but also there could be some q accessible at p such that $q' \vDash A$ for all q' accessible at q, meaning it very well could be the case that $p \nvDash \Box(\Box A \vee \Box{\sim}\Box A)$.

Exercise 3.2

(1) First we show that $\Box X \supset X$ is $S4$ valid. Consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. We must show that $p \vDash \Box X \supset X$. So suppose that $p \vDash \Box X$. Then, for each q accessible from p, $q \vDash X$. But, because $\mathcal{M}$ is an $S4$ model, its accesibility relation is reflexive, meaning p is accessible from p, and thus $p \vDash X$. Hence $p \vDash \Box X \supset X$, meaning $\Box X \supset X$ is $S4$ valid.

(2) Next we show that $\Box X \supset \Box\Box X$ is $S4$ valid. Consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. We must show that $p \vDash \Box X \supset \Box\Box X$. So suppose that $p \vDash \Box X$. Then, for each q accessible from p, $q \vDash X$. Also, for each q' accessible from some q accessible from p, we have $q' \vDash X$, because $\mathcal{M}$ is $S4$, meaning its accessibility relation is transitive. But this means $p \vDash \Box\Box X$, hence $p \vDash \Box X \supset \Box\Box X$, and thus $\Box X \supset \Box\Box X$ is $S4$ valid.

Exercise 3.3

We show that $\Diamond X \supset \Box\Diamond X$ is not $S4$ valid. Consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. If $p \vDash \Diamond X$, all we know is that $q \vDash X$ for some q accessible from p. This does not mean that $q' \vDash X$ for some q' accessible from q for each q accessible from p, which is what we would need to know for $p \vDash \Box\Diamond X$.

Exercise 3.4

(1) We first show the $S4$ validity of $\Box\Box X \equiv \Box X$. The first part, showing that $\Box X \supset \Box\Box X$ is $S4$ valid, was done in Exercise 3.2. Thus we only need to show that $\Box\Box X \supset \Box X$ is $S4$ valid. So consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. Suppose $p \vDash \Box\Box X$. Thus $q' \vDash X$ for each q' accessible from each q accessible from p. Well, q is accessible from q for each q accessible from p, because $\mathcal{M}$'s accessibility relation is reflexive. Thus $q \vDash X$ for each q accessible from p, meaning $p \vDash \Box X$, and consequently, $p \vDash \Box\Box X \supset \Box X$, and thus $\Box\Box X \supset \Box X$ is $S4$ valid.

(2) Next, we show that $\Diamond\Diamond X \equiv \Diamond X$ is $S4$ valid. Consider any $S4$ model $\mathcal{M}$, and any of its accessible worlds p. We must show that $p \vDash \Diamond\Diamond X \equiv \Diamond X$.

On the one hand, we show that $p \vDash \Diamond\Diamond X \supset \Diamond X$. So suppose that $p \vDash \Diamond\Diamond X$. Thus $q' \vDash X$ for some q' accessible from some q accessible from p. For this q', we have $p\mathcal{R}q$ and $q\mathcal{R}q'$. But this means that $p\mathcal{R}q'$ (because $\mathcal{M}$ is $S4$), and thus q' is accessible from p, meaning $p \vDash \Diamond X$.

On the other hand, we show that $p \vDash \Diamond X \supset \Diamond\Diamond X$. So suppose that $p \vDash \Diamond X$. Thus $q \vDash X$ for some q accessible from p. But q is accessible from q. Hence $q' \vDash X$ for some q' that is accessible from some q that is accessible from p (q' is itself q), meaning $p \vDash \Diamond\Diamond X$.

Section 4

Exercise 4.1

(2) We verify the $S4$ validity of the translate of

$$(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$$

Suppose that there was some $S4$ model and some possible world p in it such that

$$p \not\models [\![(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))]\!]$$

By (1) of Proposition 4.6, this would mean that there was a possible world p' accessible from p such that $p' \models [\![(A \supset (B \supset C))]\!]$ and $p' \models [\![\sim((A \supset B) \supset (A \supset C))]\!]$, which implies $p' \models \sim[\![((A \supset B) \supset (A \supset C))]\!]$ (by Proposition 4.3). By Proposition 4.5, we also have

$$(1)\ p' \models ([\![A]\!] \supset [\![(B \supset C)]\!])$$

Then again by Proposition 4.6, there is some p'' accessible from p' such that $p'' \models [\![(A \supset B)]\!]$ and $p'' \models [\![\sim(A \supset C)]\!]$, which implies that $p'' \models \sim[\![(A \supset C)]\!]$. Then we have

$$(2)\ p'' \models ([\![A]\!] \supset [\![B]\!])$$

Then again by Proposition 4.6, there is some p''' accessible from p'' such that $p''' \models [\![A]\!]$ and $p''' \models [\![\sim C]\!]$, which implies that $p''' \models \sim[\![C]\!]$. By (1) and (2), because we know that $p'\mathcal{R}p'''$ and $p''\mathcal{R}p'''$, we have

$$(1')p''' \models ([\![A]\!] \supset [\![(B \supset C)]\!])$$

$$(2')p''' \models ([\![A]\!] \supset [\![B]\!])$$

Hence, because we have $p''' \models [\![A]\!]$, (2') yields us $p''' \models [\![B]\!]$. Also (1') yields us $p''' \models [\![(B \supset C)]\!]$, and thus we have $p''' \models ([\![B]\!] \supset [\![C]\!])$, and because we have $p''' \models [\![B]\!]$, we therefore have $p''' \models [\![C]\!]$. But we also have $p''' \models \sim[\![C]\!]$, which is a contradiction. Hence it could not be that

$$p \not\models [\![(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))]\!]$$

and, consequently,

$$[\![(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))]\!]$$

is $S4$ valid.

(3) We verify the $S4$ validity of the translate of

$$(\sim A \supset \sim B) \supset (B \supset A)$$

Suppose that there was some $S4$ model and some possible world p in it such that

$$p \not\models [\![(\sim A \supset \sim B) \supset (B \supset A)$$

By (1) of Proposition 4.6, this would mean that there was a possible world p' accessible from p such that $p' \models [\![\sim A \supset \sim B]\!]$ and $p' \models [\![\sim(B \supset A)]\!]$, which implies $p' \models \sim[\![B \supset A]\!]$. Then we have

$$(1)\ p' \models [\![\sim A]\!] \supset [\![\sim B]\!]$$

Then again by Proposition 4.6, there is some p'' accessible from p' such that $p'' \models [\![B]\!]$ and $p'' \models [\![\sim A]\!]$. Because p'' is accessible from p', it follows that

$$(1')\ p'' \models [\![\sim A]\!] \supset [\![\sim B]\!]$$

Hence, by (1'), and because we have $p'' \models [\![\sim A]\!]$, we have $p'' \models [\![\sim B]\!]$, which implies $p'' \models \sim[\![B]\!]$. But we also have $p'' \models [\![B]\!]$, a contradiction. Hence it could not be that

$$p \not\models [\![(\sim A \supset \sim B) \supset (B \supset A)$$

and, consequently,

$$[\![(\sim A \supset \sim B) \supset (B \supset A)$$

is $S4$ valid.

(4) (Note that I did not understand why we need the requirenment that y is a variable such that x does not occur in the scope of $(\forall y)$ in $\phi(x)$). We verify the $S4$ validity of the translate of

$$(\forall x)\phi(x) \supset \phi(y)$$

Suppose that there was some $S4$ model, some possible world p in it, and some $d \in \mathcal{D}$, such that

$$p \not\models [\![(\forall x)\phi(x) \supset \phi(d)]\!]$$

By (1) of Proposition 4.6, this would mean that there was a possible world p' accessible from p such that $p' \models [\![(\forall x)\phi(x)]\!]$ and $p' \models [\![\sim\phi(d)]\!]$, meaning $p' \models \sim[\![\phi(d)]\!]$. Thus (by Proposition 4.5) $p' \models (\forall x)[\![\phi(x)]\!]$, meaning $p' \models [\![\phi(d)]\!]$, a contradiction. Thus it cannot be that

$$p \not\models [\![(\forall x)\phi(x) \supset \phi(d)]\!]$$

and, consequently, $[\![(\forall x)\phi(x) \supset \phi(d)]\!$ is $S4$ valid.

(5) We verify the $S4$ validity of the translate of

$$(\forall x)(\psi \supset \phi(x)) \supset (\psi \supset (\forall x)\phi(x))$$

where x does not occur free in ψ. Suppose that there was some $S4$ model and some possible world p in it such that

$$p \not\models [\![(\forall x)(\psi \supset \phi(x)) \supset (\psi \supset (\forall x)\phi(x))]\!]$$

By (1) of Proposition 4.6, this would mean that there was a possible world p' accessible from p such that $p' \models [\![(\forall x)(\psi \supset \phi(x))]\!]$ and $p' \models [\![\sim(\psi \supset (\forall x)\phi(x))]\!]$. Again by Proposition 4.6, there is some possible world p'' accessible from p' such that $p'' \models [\![\psi]\!]$ and $p'' \models [\![\sim(\forall x)\phi(x)]\!]$, which is the same as saying $p'' \models [\![(\exists x)\sim\phi(x)]\!]$. By Proposition 4.6 (4), there is some p''' accessible from p'' such that $p''' \models (\exists x)[\![\sim\phi(x)]\!]$, which implies $p''' \models (\exists x)\sim[\![\phi(x)]\!]$. This means that, for some $d \in \mathcal{D}$, $p''' \models \sim[\![\phi(d)]\!]$. Because the model is $S4$, we know that p''' is accessible from p' and p'', and thus $p''' \models [\![(\forall x)(\psi \supset \phi(x))]\!]$ and $p''' \models [\![\psi]\!]$. By Proposition 4.5, we then also have $p''' \models (\forall x)[\![\psi \supset \phi(x)]\!]$, and thus also $p''' \models [\![\psi \supset \phi(d)]\!]$ (ψ is unchanged because we assumed x does not occur free in ψ). Then, by Proposition 4.5, we also have $p''' \models ([\![\psi]\!] \supset [\![\phi(d)]\!])$, which means, because we have $p''' \models [\![\psi]\!]$, we also have $p''' \models [\![\phi(d)]\!]$. But we have $p''' \models \sim[\![\phi(d)]\!]$, a contradiction. Thus it cannot be that

$$p \not\models [\![(\forall x)(\psi \supset \phi(x)) \supset (\psi \supset (\forall x)\phi(x))]\!]$$

and, consequently,

$$[\![(\forall x)(\psi \supset \phi(x)) \supset (\psi \supset (\forall x)\phi(x))]\!]$$

is $S4$ valid.

Exercise 4.2

We show that $\Box\Diamond X \supset \Box\Diamond\Box\Diamond X$ is $S4$ valid. Well, $X \supset \Diamond X$ is $S4$ valid. Thus $\Box\Diamond X \supset \Diamond\Box\Diamond X$ is $S4$ valid. Thus, by Proposition 2.6, $\Box\Box\Diamond X \supset \Box\Diamond\Box\Diamond X$ is $S4$ valid. But $\Box X \supset \Box\Box X$ is $S4$ valid. Hence $\Box\Diamond X \supset \Box\Diamond\Box\Diamond X$ is $S4$ valid.

Exercise 4.3

(1) First, we prove that (2) of Lemma 4.2 is $S4$ valid, that is,

$$\Box\Diamond(\Box X \supset \Box\Diamond Y) \supset (\Box X \supset \Box\Diamond Y)$$

We do this by verifying the list of implications given in part (2) of the proof of Lemma 4.2.

$$\Box\Diamond(\Box \supset \Box\Diamond Y) \supset \Box\Diamond(\sim\Box X \vee \Box\Diamond Y)$$

follows by the definition of $\supset$. Next,

$$\Box\Diamond(\sim\Box X \vee \Box\Diamond Y) \supset \Box(\Diamond\sim\Box X \vee \Diamond\Box\Diamond Y)$$

is valid by Exercise 3.4 and the Replacement Theorem. Next,

$$\Box(\Diamond\sim\Box X \vee \Diamond\Box\Diamond Y) \supset \Box(\sim\Box\Box X \vee \Diamond\Box\Diamond Y)$$

follows by the definition of $\Diamond$. Next,

$$\Box(\sim\Box\Box X \vee \Diamond\Box\Diamond Y) \supset \Box(\Box\Box X \supset \Diamond\Box\Diamond)$$

follows by the definition of $\supset$. Next,

$$\Box(\Box\Box X \supset \Diamond\Box\Diamond) \supset \Box\Box\Box X \supset \Box\Diamond\Box\Diamond Y$$

Follows by the K axiom. Lastly,

$$(\Box\Box\Box X \supset \Box\Diamond\Box\Diamond Y) \supset (\Box X \supset \Box\Diamond Y)$$

follows by using the Transitivity axiom as well as part (1) of Lemma 4.2.

(2) Next, we prove that the two formulas listed in this exercise are $S4$ valid.

(a) First, we show that

$$\Box\Diamond(\Box\Diamond X \wedge \Box\Diamond Y) \supset (\Box\Diamond X \wedge \Box\Diamond Y)$$

is $S4$ valid. Now,

$$\Box\Diamond(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond\Box\Diamond X$$

is of course valid, and likewise

$$\Box\Diamond(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond\Box\Diamond Y$$

Using part (1) of Lemma 4.2, we then get

$$\Box\Diamond(\Box\Diamond X \wedge \Box\Diamond Y) \supset (\Box\Diamond X \wedge \Box\Diamond Y)$$

being $S4$ valid.

(b) Next, we show that

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond(\Box\Diamond Y \wedge \Box\Diamond Y)$$

is $S4$ valid. Well, $X \supset \Diamond X$ is $S4$ valid, meaning

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Diamond(\Box\Diamond Y \wedge \Box\Diamond Y)$$

is $S4$ valid. Then we have (by the K axiom)

$$\Box(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond(\Box\Diamond Y \wedge \Box\Diamond Y)$$

being $S4$ valid. Next we show that

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box(\Box\Diamond X \wedge \Box\Diamond Y)$$

is $S4$ valid. First,

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond X$$

and

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond Y$$

are both $S4$ valid (by Exercise 2.5(a)). Also,

$$\Box\Diamond X \supset \Box\Box\Diamond X \text{ and } \Box\Diamond Y \supset \Box\Box\Diamond Y$$

are both $S4$ valid (by the Transitivity axiom). Thus we have

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset (\Box\Box\Diamond X \wedge \Box\Box\Diamond Y)$$

being $S4$ valid, and

$$(\Box\Box\Diamond X \wedge \Box\Box\Diamond Y) \supset \Box(\Box\Diamond X \wedge \Box\Box\Diamond Y)$$

is $S4$ valid by Exercise 2.5(b), meaning

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box(\Box\Diamond X \wedge \Box\Box\Diamond Y)$$

is $S4$ valid. Hence

$$(\Box\Diamond X \wedge \Box\Diamond Y) \supset \Box\Diamond(\Box\Diamond Y \wedge \Box\Diamond Y)$$

is $S4$ valid.

Exercise 4.4

(a) We first prove part (4) of Proposition 4.3. Consider some $\sim X$. Then $[\![\sim X]\!] = \Box\Diamond\sim[\![X]\!]$. We must show that this implies $\sim[\![X]\!]$. Well, $[\![X]\!] = \Box\Diamond Y$ for some Y (by definition of the translation). Thus $[\![\sim X]\!] = \Box\Diamond\sim\Box\Diamond Y$ and $\sim[\![X]\!] = \sim\Box\Diamond Y$. To show that

$$\Box\Diamond\sim\Box\Diamond Y \supset \sim\Box\Diamond Y$$

is $S4$ valid, consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. We must show that

$$p \vDash \Box\Diamond\sim\Box\Diamond Y \supset \sim\Box\Diamond Y$$

So suppose that $p \vDash \Box\Diamond\sim\Box\Diamond Y$. Hence, for every q accessible from p, $q \vDash \Diamond\sim\Box\Diamond Y$. Thus, for every q accessible from p, there is some p' accessible from q such that $p' \vDash \sim\Box\Diamond Y$, and for every such p', there is some q' accessible from p' such that $q' \vDash \sim\Diamond Y$. But because the accessibility relation is transitive, this implies that any such q' is accessible from p. Hence $p \vDash \sim\Box\Diamond Y$, that is, $p \vDash \sim[\![X]\!]$.

Note that the converse of this problem does not hold, that is, $\sim[\![X]\!] \supset [\![\sim X]\!]$. For consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. Suppose that $p \vDash \sim[\![X]\!]$. Then, for some Y (such that $[\![X]\!] = \Box\Diamond Y$), (1) $p \vDash \sim\Box\Diamond Y$. We are trying to see if this implies that (2) $p \vDash \Box\Diamond\sim\Box\Diamond Y$. However, (2) means that, for every q' accessible from p, there is some r accessible from q' such that (3) $r \vDash \sim\Box\Diamond Y$. But (1) does not imply this, as, if we only knew that, there could be some q' accessible from p where there is no r accessible from q' such that (3) holds for r.

(b) Next, we prove part (5) of Proposition 4.3. Now, $(\exists x)[\![\phi(x)]\!] = (\exists x)\Box\Diamond Y$ for some Y. Also,

$$[\![(\exists x)\phi(x)]\!] = \Box\Diamond(\exists x)\Box\Diamond Y$$

We prove the result directly, by considering some $S4$ model $\mathcal{M}$, and any of its possible worlds p. Suppose that

$$(1)\ p \vDash (\exists x)\Box\Diamond Y$$

We must show that

$$(2)\ p \vDash \Box\Diamond(\exists x)\Box\Diamond Y$$

What (1) says is that for some $d \in \mathcal{D}$, for every world q accessible from p, there is some world r accessible from q such that $r \vDash Y(d)$ (where $Y(d)$ means what is expected if Y is atomic, whereas, if Y is of the form $[\![\psi(x)]\!]$, then $Y(a)$ is $[\![\psi(d)]\!]$). What (2) says is that for every q accessible from p, there is some r accessible from q, such that there is some $d \in \mathcal{D}$, such that for every q' accessible from r, there is some r' accessible from q' such that $r' \vDash Y(d)$. But any such q' is accessible from p (by the transitivty of the accessiblity relation), meaning there will always be some such r' for every q', meaning (2) follows from (1).

Note that the converse of (5) does not follow, that is,

$$[\![(\exists x)\phi(x)]\!] \supset (\exists x)[\![\phi(x)]\!]$$

For consider any $S4$ model $\mathcal{M}$ and any of its possible worlds p. For some Y,

$$(\exists x)[\![\phi(x)]\!] = (\exists x)\Box\Diamond Y$$

and

$$[\![(\exists x)\phi(x)]\!] = \Box\Diamond(\exists x)\Box\Diamond Y$$

So suppose that $p \Vdash [\![(\exists x)\phi(x)]\!]$, that is,

$$(1)\ p \Vdash \Box\Diamond(\exists x)\Box\Diamond Y$$

We are to show that

$$(2)\ p \Vdash (\exists x)\Box\Diamond Y$$

does not necessarily follow. Well, (1) holding means that, for every q accessible from p, there is some r accessible from q such that, for some $d \in \mathcal{D}$, for every q' accessible from r, there is some r' accessible from q' such that $r' \vDash Y(d)$ (call such an r' good). What (2) says is that there is some $d \in \mathcal{D}$ such that for every q accessible from p, there is some r accessible from q such that $r \vDash Y(d)$. But this is not implied by (1). We know that there is some $d \in \mathcal{D}$ *for each* good r', but this d may differ from r' to r', whereas the d stays constant in (2).

Exercise 4.5

We prove part (2) of Proposition 4.4, that is, that

$$[\![X \supset Y]\!] \equiv \Box\Diamond([\![X]\!] \supset [\![Y]\!])$$

is $S4$ valid. By the definition of the connective $\supset$, $[\![X \supset Y]\!]$ is $[\![\sim(X \wedge \sim Y)]\!]$, and by the definition of the translation, this is equivalent to $\Box\Diamond\sim[\![X \wedge \sim Y]\!]$. By part (3) of Proposition 4.3 (and the replacement theorem), this is equivalent to $\Box\Diamond\sim([\![X]\!] \wedge [\![\sim Y]\!])$, which is in turn equivalent to $\Box\Diamond\sim([\![X]\!] \wedge \Box\Diamond\sim[\![Y]\!])$. By part (1) of Proposition 4.3, this is equivalent to $\Box\Diamond\sim(\Box[\![X]\!] \wedge \Box\Diamond\sim[\![Y]\!])$, which, by the distributivity of $\Box$, is in turn equivalent to $\Box\Diamond\sim\Box([\![X]\!] \wedge \Diamond\sim[\![Y]\!])$. Now, by definition of $\Diamond$, this is equivalent to $\Box\Diamond\Diamond\sim([\![X]\!] \wedge \sim\Box[\![Y]\!])$, which, by Proposition 4.3 (1), is equivalent to $\Box\Diamond\sim\Box([\![X]\!] \wedge \sim[\![Y]\!])$. Now, this is equivalent to $\Box\Diamond\sim([\![X]\!] \wedge \sim\Box[\![Y]\!])$ (because $\Diamond\Diamond X \equiv \Diamond X$), and this is the same as saying $\Box\Diamond([\![X]\!] \supset [\![Y]\!])$.

Here is a note on part (3) of Proposition 4.4. Here is a chain of formulas, each one equivalent to the next, which gives more explicit detail why $\Box\Diamond\sim\Box\Diamond(\exists x)\Box\Diamond\sim[\![\phi]\!]$ is equivalent to $\Box\Diamond\Diamond\Box(\forall x)\Diamond\Box[\![\phi]\!]$:

$\Box\Diamond\sim\Box\Diamond(\exists x)\Box\Diamond\sim X$
$\Box\Diamond\sim\Box\Diamond\sim(\forall x)\sim\Box\Diamond\sim X$
$\Box\Diamond\sim\Box\Diamond\sim(\forall x)\Diamond\sim\Diamond\sim X$
$\Box\Diamond\sim\Box\Diamond\sim(\forall x)\Diamond\Box X$
$\Box\Diamond\Diamond\sim\Diamond\sim(\forall x)\Diamond\Box$
$\Box\Diamond\Diamond\Box(\forall x)\Diamond\Box$

Exercise 4.6

(1) Here is why part (1) of Proposition 4.6 is true. Suppose $p \vDash \sim[\![X \supset Y]\!]$. Then $p \vDash \sim\Box\Diamond([\![X]\!] \supset [\![Y]\!])$, which is equivalent to $p \vDash \Diamond\Box\sim([\![X]\!] \supset [\![Y]\!])$, which is equivalent to $p \vDash \Diamond\Box\sim(\Box[\![X]\!] \supset \Box\Diamond[\![Y]\!])$. Thus, for some q accessible from p, $q \vDash \Box\sim(\Box[\![X]\!] \supset \Box\Diamond[\![Y]\!])$. Hence, from all r accessible from q, $r \vDash \sim(\Box[\![X]\!] \supset \Box\Diamond[\![Y]\!])$, which is the same as saying $r \vDash \Box[\![X]\!]$ and $r \vDash \sim\Box\Diamond[\![Y]\!]$, that is, $r \vDash \Diamond\Box\sim[\![Y]\!]$. Thus, for some p' accessible from r, $p' \vDash [\![X]\!]$ and $p' \vDash \Box\sim[\![Y]\!]$ (and this implies $p' \vDash \Box\Diamond\sim[\![X]\!]$, that is, $p' \vDash [\![\sim X]\!]$). Because we are considering an $S4$ model, it follows by the transitivity of the accessibility relation that all such p' are accessible from p, and therefore we are done.

(2) Here is why part (3) of Proposition 4.6 is true. Suppose $p \vDash [\![X \vee Y]\!]$. Then $p \vDash \Box\Diamond([\![X]\!] \vee [\![Y]\!])$, meaning for every q accessible from p, there is some r accessible from q such that $r \vDash [\![X]\!] \vee [\![Y]\!]$. But, because we are considering an $S4$ model, it follows by the transitivity of the accessibility relation that all such r are accessible from p.

(3) Here is why part (4) of Proposition 4.6 is true. Suppose $p \vDash [\![(\exists x)\phi(x)]\!]$. Then $p \vDash \Box\Diamond(\exists x)[\![\phi(x)]\!]$, meaning for every q accessible from p, there is some r accessible from q such that $r \vDash (\exists x)[\![\phi(x)]\!]$. But, because we are considering an $S4$ model, it follows by the transitivity of the accessibility relation that all such r are accessible from p.

Exercise 4.7

(a) This formula is not $S4$ valid. For suppose $p \vDash [\![X \vee Y]\!]$, which is equivalent to (1) $p \vDash \Box\Diamond([\![X]\!] \vee [\![Y]\!])$ (by part (2) of Proposition 4.3). We need to show that $p \vDash [\![X]\!] \vee [\![Y]\!]$, that is, (2) $p \vDash \Box\Diamond[\![X]\!] \vee \Box\Diamond[\![Y]\!]$, does not necessarily follow. Well, what (1) means is that, for every q accessible from p, there is some r accessible from q such that either $r \vDash [\![X]\!]$ or $r \vDash [\![Y]\!]$. But this doesn't meet the conditions of (2); it could indeed be that there was some q accessible from p such that there was no r accessible from q such that $r \vDash [\![X]\!]$, and that there was some q accessible from p such that there was no r accessible from q such that $r \vDash [\![Y]\!]$.

(b) This formula is just the formula listed in part (2) of Proposition 4.5. It is claimed there that the proof of that part is *similar* to that of part (1), though I cannot see why that is. Here is the proof that I found. Suppose $p \vDash [\![X]\!] \vee [\![Y]\!]$, which is equivalent to $p \vDash \Box\Diamond[\![X]\!] \vee \Box\Diamond[\![Y]\!]$ by (2) of Proposition 4.3. Thus, it is either the case that, (1) for every q accessible from p, there is some r accessible from q such that $r \vDash [\![X]\!]$, or it is the case that, (2) for every q accessible from p, there is some r accessible from q such that $r \vDash [\![Y]\!]$. By part (1) of Proposition 4.4, it suffices to show that (3) $p \vDash \Box\Diamond([\![X]\!] \vee [\![Y]\!])$. Well, regardless of which of (1) or (2) is true, (3) certainly follows, as for every q accessible from p, there is some r accessible from q such that either $r \vDash [\![X]\!]$ or $r \vDash [\![Y]\!]$.

Also note that the converse of part (1) of Proposition 4.5 is not $S4$ valid.

Exercise 4.8

We show that for any closed formula X in the language of classical logic, a finite set S of classical formulas logically implies X iff $[\![X]\!]$ is logically implied the set containing the translates of all the formulas in S.

On the one hand, suppose that a finite set S of classical formulas logically implies a classical formula X. Suppose the formulas in S are $S_1, ..., S_n$. Thus $(S_1 \wedge ... \wedge S_n) \supset X$ is classically valid. By Proposition 4.7, it follows that $[\![(S_1 \wedge ... \wedge S_n) \supset X]\!]$ is valid over all $S4$ models. Using part (3) of Proposition 4.3, one can see that this means $[\![([\![S_1]\!] \wedge ... \wedge [\![S_n]\!]) \supset [\![X]\!]]\!]$ is valid over all $S4$ models, which is what we were required to show.

On the other hand, suppose that X is not logically implied by some finite set S (say it contains the formulas $S_1, ..., S_n$) of formulas in some classical model M. Construct an $S4$ model $(\mathcal{G}, \mathcal{R}, \mathcal{D}, \mathcal{V})$, as follows. Let $\mathcal{G}$ be $\{p\}$ where p is some arbitrary set. Set $p\mathcal{R}p$. Take for $\mathcal{D}$ the domain of the classical model M. And, for an atomic formula A, set $p \vDash A$ iff A is true in M. In this model, for any formula Z, $p \vDash Z$ iff $p \vDash \Box Z$ iff $p \vDash \Diamond Z$. Then, by induction on degrees, it follows that for each closed formula Z, Z is true in M iff $p \vDash [\![Z]\!]$. Since X is not logically implied by S in M, then $(S_1 \wedge ... \wedge S_n) \supset X$ is not true over M, meaning $p \nvDash [\![(S_1 \wedge ... \wedge S_n) \supset X]\!]$, and, using part (3) of Proposition 4.3, it follows that $p \nvDash [\![([\![S_1]\!] \wedge ... \wedge [\![S_n]\!]) \supset [\![X]\!])]\!]$. Contrapositively, if $([\![S_1]\!] \wedge ... \wedge [\![S_n]\!]) \supset [\![X]\!]$ is valid over all $S4$ models, $(S_1 \wedge ... \wedge S_n) \supset X$ is valid over all classical models (X is logically implied by S).